

Nathan (Kaiyan) Zhang

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Updated August 2026

Education

The University of British Columbia

Sep 2023 – May 2028

B.A., Combined Major in Economics and Statistics; Minor in Data Science; Vancouver, BC

- Cumulative average: 93.5/100

Research Interests

Methods: Statistical learning and inference; causal inference and econometrics; panel and time-series analysis; text and image methods.

Applications: Technological change and labor; financial risk and ESG; computational social science.

Research Experience

Research Assistant, UBC Sauder School of Business

Apr 2026 – Present

Supervisors: J. Frank Li and Cheng Chen

- Translated a faculty-developed model of age-biased technological change into an empirical pipeline linking CPS, PSID, NLSY, O*NET, and PATSTAT. As the project's sole RA, I built the worker samples, occupation crosswalks, and wage and sorting models.
- Harmonized 60+ O*NET releases, extracted task relations with NER/RE, and compared task text with about 56 million PATSTAT records.
- Implemented share and log-share GMM, a 999-replication common-year block bootstrap, calibration, and robustness checks. I also drafted the data section and maintain the R replication package.

Research Assistant, UBC Sauder School of Business

May 2026 – Present

Supervisor: Allen Hu

- Reconciled 1,330 RavenPack publication values against 3,210,045 ProQuest articles dated July 1861 through May 2026. The audit covers missingness, provenance, and field reliability before text analysis.
- Built the collection and validation pipeline for an 85,259-ID financial-creator directory. It preserves raw media and metadata, run manifests, evidence links, and reconciliation results.

Research Assistant, Vancouver School of Economics at UBC

Jan 2025 – May 2026

Supervisors: Jonathan Graves and Laura Nelson

- Built and maintained Python/R teaching modules for COMET and Praxis UBC across panel econometrics, forecasting, NLP, and computer vision. I also maintained the Quarto/GitHub Actions publishing pipeline used to serve 1,000+ cumulative student users.
- Designed, hosted, and taught a computer vision workshop for social science research. I also coauthored a Programming Historian tutorial under site-maintainer review, prepared APSA workshop materials, and provided on-site technical support.

Manuscripts

Memory Is the Difference between Two Futures: Memory Response Paths and Path-Structured Context 2026 Compilation

Manuscript under review; with Jerrold Huang

Defines a memory's effect as the difference between matched futures with and without it, traced over four horizons in 180 independent ecologies. In a separate 144-item evaluation, PSCC's exact-match accuracy is 6.94 percentage points higher than A-Mem's (95% CI: 2.64 to 11.32) while using 67.2% of its API tokens.

When Answers Surprise

2026

Manuscript under review; with Jerrold Huang

Tests whether the novelty of a management answer predicts returns beyond earnings surprise. The panel links about 31,000 deduplicated 2018-2025 firm events to Capital IQ transcripts, IBES, CRSP, and historical S&P 500 membership. Evaluation uses time holdouts, learner checks, placebos, and timing checks.

Selected Projects

Double Machine Learning for Payment-Friction Spending Effects 2025

Course research project [Code] [Report]

- Used cross-fitted DoubleML on 5,583 respondents and 22,332 in-person person-days. Under selection on observables and overlap, the estimated conditional effect of lower-friction payment mixes on daily spending is about 5%, with heterogeneity by ticket size, income, and education.

Canadian Inflation Forecasting and Macro Signal Evaluation 2025

Forecasting project [Code]

- Built an R/Quarto VAR forecast for Canadian inflation. Rolling out-of-sample tests against naive and random-walk baselines yield an RMSE of 0.16 percentage points. Presented impulse responses in a Bank of Canada-style policy briefing.

Teaching and Research Infrastructure

Image Analysis Workshop: Computer Vision for Social Science Research Mar 25, 2026

Workshop lead and instructor [Workshop]

Designed, hosted, and taught a Praxis UBC workshop that moved from preprocessing and BoVW to CNN/ViT embeddings, classification, object detection, OCR, and captioning.

A Study of Richter's Kouroi Through Image Embedding Jul 25, 2025

Coauthor with Yash Mali and Krishaant Pathman [Notebook]

Used ConvNeXt V2 embeddings, PCA, clustering, and classification to test whether image embeddings recover structure in Richter's kouroi categories.

LLM Prediction Module 2025

Coauthor [Module]

Coauthored course-adopted material connecting next-word prediction, token probabilities, and economics examples.

Professional Experience

Incoming ESG Co-op, Capital Markets Sep 2026 – Dec 2026

British Columbia Investment Management Corporation (BCI); Vancouver, BC

Teaching Assistant Sep 2024 – Aug 2025

UBC Chinese Language Program; Vancouver, BC

- Supported two Chinese language courses through grading, classroom support, volunteer training, and feedback coordination.

Awards and Scholarships

Work Learn International Undergraduate Research Awards (WLIURA) (UBC) 2026

Economics Undergraduate Scholarship (Vancouver School of Economics, UBC) 2025

Faculty of Arts International Student Scholarship (UBC Faculty of Arts) 2024, 2025

Selected Coursework

Statistical Inference (100); Machine Learning and Data Mining (93); Causal Methods (98); Advanced Econometrics (98); Statistical Learning (97); Time Series and Forecasting (95)

Technical Skills and Languages

Programming and data: Python, R, SQL, Julia, DuckDB, Polars, Git/GitHub Actions, HPC/Slurm, Jupyter/Quarto

Modeling: causal inference/econometrics, panel/time series, machine learning, NLP, computer vision, scikit-learn, PyTorch, Transformers, fixest, statsmodels

Languages: Mandarin Chinese: native; English: academic working proficiency; Japanese: JLPT N3